Supplementary Material

[final EML title]

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S1 Model derivation

State variables and kinematics

Following Klempt et al. (2026, 2024), the biofilm is described at a material point by two sets of internal variables: the volume fraction $\phi_i \in [0,1]$ occupied by species i ($i = 1,\dots,n$; here $n = 5$), and the viability fraction $\psi_i \in [0,1]$ representing the proportion of living cells within species i. The living volume fraction is then

$$\bar{\phi}_i := \phi_i\,\psi_i, \qquad i = 1,\dots,n. \tag{S1}$$

An additional void fraction ϕ_0 closes the system via the holonomic constraint

$$\sum_{l=0}^{n} \phi_l(t) - 1 = 0 \quad \forall t. \tag{S2}$$

Variational formulation

The evolution equations are derived from the principle of the minimum of the dissipation potential (Junker et al., 2014; Hackl, 1997), the local form of the extended Hamilton principle (Junker and Balzani, 2021; Junker et al., 2025):

$$\dot{\Psi} + \Delta + C \to \mathrm{stat}_{(\dot{\xi},\gamma)}, \tag{S3}$$

where $\xi = (\phi,\psi)^T$ collects the volume fractions and viability fractions, γ is a Lagrange multiplier enforcing the volume constraint, C is the constraint function, and Δ the dissipation function. The free energy density encodes nutrient-driven growth and antibiotic-induced killing:

$$\Psi = -\tfrac{1}{2}\,c^*\,\bar{\phi}^T\mathbf{A}\,\bar{\phi} + \tfrac{1}{2}\,\alpha^*\,\psi^T\mathbf{B}\,\psi, \tag{S4}$$

where c^* is the nutrient concentration, α^* the antibiotic concentration, $\mathbf{A} \in \mathbb{R}^{n\times n}$ the symmetric growth coefficient matrix, and $\mathbf{B} = \mathrm{diag}(b_1,\dots,b_n)$ the antibiotic sensitivity matrix.

The dissipation function is derived from the rate-dependent potential (Klempt et al., 2026):

$$\Delta = \tfrac{1}{2}\,\dot{\bar{\phi}}^T\eta\,\dot{\bar{\phi}} + \tfrac{1}{2}\,\dot{\phi}^T\eta\,\dot{\phi}, \tag{S5}$$

with the diagonal viscosity matrix $\eta = \mathrm{diag}(\eta_1,\dots,\eta_n)$. The first term couples $\dot{\phi}_i$ and $\dot{\psi}_i$ through $\dot{\bar{\phi}}_i = \dot{\phi}_i\psi_i + \phi_i\dot{\psi}_i$, giving rise to non-linear dissipative interactions; the second term regularises the system to full rank. Following Klempt et al. (2026), the viscosity parameters are set to $\eta_i = 1$ for all species, rendering them dimensionless scaling factors; the physical time scale is absorbed into the time-step size Δt.

The volume constraint (S2) enters through a Lagrange multiplier γ:

$$C = \gamma\Big(\sum_{l=0}^{n}\phi_l - 1\Big). \tag{S6}$$

Stationarity conditions

Evaluating the stationarity conditions yields the coupled evolution equations (Klempt et al., 2026):

Volume fraction ($i = 1, \ldots, n$):

$$0 = -c^* \, \psi_i \left(\mathbf{A}\bar{\phi}\right)_i + \eta_i \left(\dot{\phi}_i \, \psi_i^2 + \bar{\phi}_i \, \dot{\psi}_i + \dot{\phi}_i\right) + \gamma, \tag{S7}$$

Viability fraction ($i = 1, \ldots, n$):

$$0 = -c^* \, \phi_i \left(\mathbf{A}\bar{\phi}\right)_i + \alpha^* \, b_i \, \psi_i + \eta_i \left(\dot{\psi}_i \, \phi_i^2 + \bar{\phi}_i \, \dot{\phi}_i\right) + \gamma, \tag{S8}$$

Void fraction:

$$0 = \gamma + \dot{\phi}_0, \tag{S9}$$

Volume constraint:

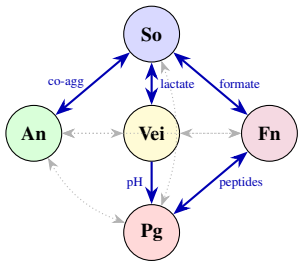
$$0 = \sum_{l=0}^{n} \phi_l - 1, \tag{S10}$$

In the numerical implementation, a logarithmic barrier penalty Ψ_{bar} with $K_p = \eta_i \times 10^{-4}$ is added to enforce the box constraints $\phi_i, \psi_i \in (0,1)$ (Klempt et al., 2026); its contribution is omitted from Eqs. (S7)–(S9) for clarity. The system (S7)–(S10) constitutes $2n+2$ coupled non-linear equations for the unknowns $(\phi_1, \ldots, \phi_n, \phi_0, \psi_1, \ldots, \psi_n, \gamma)$, solved at each time step via Newton's method with an implicit Euler discretisation ($\Delta t = 10^{-4}$, $N_{\text{step}} = 2{,}500$ steps, $t_{\max} = 0.25$ corresponding to the 21-day experimental window).

The variational derivation automatically ensures symmetry $A_{ij} = A_{ji}$ and thermodynamic consistency (Junker and Balzani, 2021). Since c^* enters the evolution equations only through the product $c^* A_{ij}$, its absolute value is non-identifiable; we fix $c^* = 25$ so that the inferred A_{ij} remain $\mathcal{O}(1)$. The Heine et al. experiments do not involve antibiotic agents, so we set $\alpha^* = 0$, which eliminates the decay term $\alpha^* b_i \psi_i$ from Eq. (S8); the antibiotic sensitivity matrix $\mathbf{B}$ is therefore inactive and excluded from estimation. The symmetric interaction matrix $\mathbf{A}$ has 15 independent entries (the main text); these form the 15-dimensional parameter vector θ.

S2 Reported interactions among the five taxa

Figure S1 and Table S1 summarise interactions reported in the literature; they are used only to interpret the inferred coefficients, never as constraints.



Pair	Mechanism	Reported flow
So–An	Co-aggregation	mutual
So–Vei	Lactate exchange	So to Vei
So–Fn	Formate symbiosis	mutual
Vei–Pg	pH elevation	Vei to Pg
Fn–Pg	Peptide supply	mutual

Table S1: Previously reported or biologically supported interactions (Heine et al., 2025) used to interpret the inferred coefficients. The model coefficients a_{ij} are symmetric; the last column refers to the reported metabolite flow, not to the model. All 15 parameters are estimated; none is fixed a priori.

Figure S1: Species interaction network (Heine et al., 2025). Solid: previously reported or biologically supported interactions considered in the interpretation of the model. Dotted grey: no reported pathway. All pairs are estimated with the same prior.

Algorithm S1 Staged GPU-accelerated TMCMC (one seed; every stage is run with three seeds and checked as in the main text)

Require: Composition data $\mathbf{y}_{\text{obs}}$, viability data, N_p particles, GPU device
Ensure: MAP estimate $\hat{\theta}^{\text{MAP}}$, posterior samples, log-evidence $\ln \hat{Z}$

1: ▷ **Stage (i): composition likelihood, initial box**
2: Draw $\{\theta_j\}_{j=1}^{N_p}$ from the stage prior; set $\beta_0 \leftarrow 0$
3: **while** $\beta_m < 1$ **do**
4: Find β_{m+1} s.t. $\text{CoV}[\{w_j\}] = \delta_{\text{target}}$ via bisection
5: Compute weights $w_j \leftarrow p(\mathbf{y}_{\text{obs}} \mid \theta_j)^{\beta_{m+1}-\beta_m}$
6: Resample N_p particles $\propto w_j$ (systematic)
7: Estimate weighted covariance $\hat{\Sigma}^{(m)}$ (weighted particle covariance)
8: **for** $k = 1$ **to** K **do**
9: Propose θ'_j for all j (random-walk or differential-evolution move)
10: Evaluate all N_p likelihoods in parallel via `jax.vmap` on GPU
11: Accept/reject via MH ; reject proposals outside the box
12: **end for**
13: **end while**
14: Record MAP and posterior SD per parameter
15: ▷ **Stage (ii): composition likelihood, box mean ± 4 SD**
16: Repeat lines 2–13 on the narrowed box, starting from the previous particles
17: ▷ **Stages (iii)–(iv): add the viability channel; initial box, then mean ± 4 SD**
18: Augment likelihood with the weighted viability channel (Table S2)
19: Repeat lines 2–13 for each stage
20: **return** $\hat{\theta}^{\text{MAP}}$, posterior samples, $\ln \hat{Z}$

S3 Likelihood weights and algorithm

Weight	Value	Applies to
λ_{Pg}	5	Pg, all time points
λ_{late}	3	Days 15 and 21, all species
λ_{rare}	0.1	species with mean fraction $< 5\%$
λ_{ψ}	3.0 (DH), 2.0 (DS), 1.5 (CS, CH)	viability channel, Phase 2

Table S2: Likelihood weights.

S4 Prior boxes

Table S4 lists the uniform prior box of every interaction entry for each condition and stage. The initial box was set empirically to contain earlier posterior estimates and encodes no sign constraint. An entry was widened only where more than 20% of its posterior mass lay in the outer 5% of the range on one side, or where this fraction increased by more than 0.10 between stages with the same likelihood; the stage was then repeated. The last column gives $\ln V = \sum_k \ln(u_k - l_k)$, the log-volume of the box, which enters $\ln \hat{Z}$ and is the reason why $\ln \hat{Z}$ is not compared across conditions.

S5 Runs and convergence

Table S5 lists, for every condition and stage, the number of particles and mutation steps, the number of tempering stages, the acceptance rate, the moves per parameter, and the spread over the three seeds of

Metric	CPU (12 cores)	GPU (RTX 4090)	Speedup
Matched benchmark (DH, $N_p = 500$, $K = 10$, 9 stages):			
Per stage	2 400 s	14 s	170×
Total	21 600 s	147 s	**147**×
*Final stage (iv) (4 cond., $N_p =$ **[n]**, $K =$ **[n]**, GPU **[model]**):*			
Per condition	**[h]**†	**[h]**	**[x]**
All 4 cond.	**[h]**†	**[h]**	**[x]**

Table S3: Computational cost comparison. The matched benchmark (500 particles, identical settings) isolates the GPU speedup at about 150–170× (170× per stage, 147× in total). The final-stage times are measured per seed **[update with the final runs]**. †Extrapolated from the 500-particle CPU benchmark assuming linear scaling in N_p.

the maximum log-likelihood and of the posterior medians (criteria (i)–(iv) of the main text).

S6 Identifiability of the interaction entries

Table S6 lists, for the final stage (iv) of every condition, the median and the 90% interval of each entry, pooled over the three seeds. An entry whose posterior standard deviation is at least 0.8 times that of a uniform distribution on its box is shown in grey and excluded from the seed-agreement criterion: the data do not constrain its magnitude beyond the prior box. Among these, entries whose 90% interval excludes zero are marked s.o. (sign only), the others n.i. (not identified).

S7 Bayes factor for the *Fn–Pg* interaction

Stage (i) was repeated with $a_{45} = 0$ for all four conditions (three seeds each). Table S7 reports $\ln \hat{Z}$ of both models, $\ln B = \ln \hat{Z}_{\text{full}} - \ln \hat{Z}_{a_{45}=0}$, and the width of the a_{45} box, which sets the Occam penalty.

Dysbiotic Static. In DS, a_{45} and a_{33} trade off. With $a_{45} = 0$ the posterior moves almost entirely to the mode $a_{33} \approx -10$ (weight **[0.83–0.87]**), whereas the full model places most of its mass at $a_{33} \approx +1$ (weight of the mode $a_{33} < -5$: **[0.09–0.12]**). Table S8 gives a_{45} separately for each mode.

S8 Likelihood weights

Stage (i) for DH was repeated with $\lambda_{\text{Pg}} = \lambda_{\text{late}} = 1$ (1000 particles, 100 mutation steps, three seeds; all convergence criteria met). Table S9 compares it with the weighted run.

References

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Table S4: Prior boxes per condition and stage for the 15 free entries; entries widened relative to stage (i) in bold. Stage (iv) uses the stage-(iii) posterior mean ± 4 SD, clipped to the stage-(iii) box. $\ln V = \sum_k \ln(u_k - l_k)$ over the free entries.

CS

Entry	Stage (i)	Stage (ii)	Stage (iv)
a_{11}	[0, 2]	[0, 2]	± 4 SD
a_{12}	[-0.5, 2]	**[-0.177412, 2]**	± 4 SD
a_{22}	[-1, 3]	[-1, 3]	± 4 SD
a_{33}	[-1, 1.5]	**[-1, 0.951294]**	± 4 SD
a_{34}	[-1, 1]	[-1, 1]	± 4 SD
a_{44}	[-1, 1]	[-1, 1]	± 4 SD
a_{13}	[-1, 3]	**[1.16755, 3]**	± 4 SD
a_{14}	[-1, 1]	[-1, 1]	± 4 SD
a_{23}	[-1, 3]	**[-0.713859, 3]**	± 4 SD
a_{24}	[-1, 1]	[-1, 1]	± 4 SD
a_{55}	[-1, 1]	[-1, 1]	± 4 SD
a_{15}	[-1, 1]	[-1, 1]	± 4 SD
a_{25}	[-1, 1]	[-1, 1]	± 4 SD
a_{35}	[-1, 1]	[-1, 1]	± 4 SD
a_{45}	[-1, 1]	[-1, 1]	± 4 SD
$\ln V$	12.9	11.7	–

CH

Entry	Stage (i)	Stage (ii)	Stage (iv)
a_{11}	[-1, 1.5]	[-1, 1.5]	± 4 SD
a_{12}	[-1, 2.5]	**[-0.840367, 2.5]**	± 4 SD
a_{22}	[-1, 2.5]	[-1, 2.5]	± 4 SD
a_{33}	[-1, 1.5]	[-1, 1.5]	± 4 SD
a_{34}	[-1, 1]	[-1, 1]	± 4 SD
a_{44}	[-1, 1.5]	[-1, 1.5]	± 4 SD
a_{13}	[-1, 3]	**[-0.403325, 3]**	± 4 SD
a_{14}	[-1, 1]	[-1, 1]	± 4 SD
a_{23}	[-1, 3]	**[-0.846905, 3]**	± 4 SD
a_{24}	[-1, 1]	[-1, 1]	± 4 SD
a_{55}	[-1, 1.5]	[-1, 1.5]	± 4 SD
a_{15}	[-1, 1]	[-1, 1]	± 4 SD
a_{25}	[-1, 1]	[-1, 1]	± 4 SD
a_{35}	[-1, 1]	[-1, 1]	± 4 SD
a_{45}	[-1, 2]	[-1, 2]	± 4 SD
$\ln V$	14.2	14.0	–

DS

Entry	Stage (i)	Stage (ii)	Stage (iv)
a_{11}	[-1, 2]	[-1, 2]	± 4 SD
a_{12}	[-1, 1]	[-1, 1]	± 4 SD
a_{22}	[-1, 1.5]	[-1, 1.5]	± 4 SD
a_{33}	[-15, 20]	**[-15, 15.3486]**	± 4 SD
a_{34}	[-15, 20]	**[-10.4661, 11.4797]**	± 4 SD
a_{44}	[-1, 2.5]	[-1, 2.5]	± 4 SD
a_{13}	[-1, 1.5]	[-1, 1.5]	± 4 SD
a_{14}	[-1, 1]	[-1, 1]	± 4 SD
a_{23}	[-1, 3]	[-1, 3]	± 4 SD
a_{24}	[-0.5, 3]	[-0.5, 3]	± 4 SD
a_{55}	[-1, 4]	**[-1, 2.48643]**	± 4 SD
a_{15}	[-0.5, 2.5]	[-0.5, 2.5]	± 4 SD
a_{25}	[-0.5, 2.5]	[-0.5, 2.5]	± 4 SD
a_{35}	[-3, 3]	**[-2.04357, 3]**	± 4 SD
a_{45}	[-15, 20]	**[-2.92412, 7.26157]**	± 4 SD
$\ln V$	24.5	22.1	–

DH

Entry	Stage (i)	Stage (ii)	Stage (iv)
a_{11}	[0, 5]	[0, 5]	± 4 SD

Table S5: Runs of the final pipeline: particles N_p, mutation steps K, tempering stages, acceptance rate, moves per free parameter, maximum log-likelihood (best seed; spread over three seeds in parentheses) and wall-clock hours per seed.

Cond.	Stage	N_p	K	Stages	Accept.	Moves/dim	max log L (spread)	Hours
CS	(i)	1000	80	6	0.14–0.14	4.4–4.5	-5.86 (0.11)	2.9–3.0
CS	(ii)	2000	120	6	0.14–0.14	6.7–6.8	-5.67 (0.27)	4.7–5.1
CS	(iv)	5000	150	6–7	0.13–0.13	7.9–9.2	-15.81 (0.24)	7.7–9.0
CH	(i)	1000	80	5	0.14–0.14	3.8–3.9	-3.19 (0.10)	2.5–2.5
CH	(ii)	2000	150	4	0.15–0.15	5.9–6.0	-3.15 (0.08)	3.9–4.2
CH	(iv)	5000	150	7	0.13–0.13	8.9–9.0	-8.65 (0.11)	9.7–9.8
DS	(i)	1000	80	10	0.11–0.12	5.9–6.2	-0.91 (0.72)	5.2–5.4
DS	(ii)	3000	100	10	0.12–0.12	8.0–8.2	-0.58 (0.27)	5.7–6.4
DS	(iv)	18000	150	10	0.11–0.12	11.4–11.6	-6.45 (0.53)	15.8–17.6
DH	(i)	1000	80	10	0.12–0.13	6.6–6.7	-3.92 (0.51)	4.9–5.1
DH	(ii)	2000	80	8–9	0.12–0.13	5.4–6.0	-3.21 (0.31)	4.3–5.2
DH	(iv)	14000	120	12	0.10–0.10	9.9–10.1	-14.65 (0.16)	15.3–16.6

Table S6: Posterior median [5%, 95%] of the 15 entries at stage (iv), three seeds pooled. Grey: the posterior spreads over most of the box (standard deviation ≥ 0.8 times that of a uniform distribution on the box); n.i.: not identified, s.o.: sign only (the 90% interval excludes zero).

Entry	CS	CH	DS	DH
a_{11}	-3.58 [-4.9, -1.2]	-0.99 [-2.5, -0.5]	$+0.02$ [-0.9, $+1.7$] n.i.	-0.85 [-4.6, $+4.3$] n.i.
a_{12}	$+0.77$ [-0.3, $+1.8$] n.i.	-0.44 [-3.5, $+0.6$]	-0.07 [-0.9, $+0.9$] n.i.	-1.35 [-4.6, $+2.3$] n.i.
a_{22}	$+0.68$ [-0.7, $+2.2$]	-0.13 [-0.9, $+1.5$]	$+0.14$ [-0.9, $+1.3$] n.i.	-2.98 [-4.7, -0.7]
a_{33}	-6.76 [-13.4, -2.5]	-0.53 [-1.0, $+0.2$]	-0.63 [-1.5, -0.3]	-1.28 [-2.8, -0.7]
a_{34}	-0.29 [-0.9, $+0.8$] n.i.	-0.20 [-0.9, $+0.8$] n.i.	-2.25 [-12.3, $+0.5$]	-1.29 [-4.4, $+3.7$]
a_{44}	-0.01 [-0.9, $+0.9$] n.i.	$+0.19$ [-0.9, $+1.4$] n.i.	-0.12 [-0.9, $+2.0$] n.i.	-6.88 [-9.7, -2.4]
a_{13}	$+5.74$ [$+2.8$, $+9.2$]	$+0.61$ [$+0.0$, $+1.4$]	-0.16 [-0.9, $+0.7$]	-1.00 [-2.7, $+1.1$]
a_{14}	-0.40 [-0.9, $+0.6$] n.i.	-2.89 [-4.8, -0.2] s.o.	-0.10 [-0.9, $+0.9$] n.i.	-3.19 [-7.0, $+1.3$] n.i.
a_{23}	$+0.50$ [-0.8, $+2.5$] n.i.	$+0.98$ [-0.1, $+2.0$]	-1.84 [-4.5, -0.1]	$+1.18$ [-2.1, $+2.9$]
a_{24}	-0.22 [-0.9, $+0.8$] n.i.	-0.13 [-0.9, $+0.8$] n.i.	$+1.16$ [-0.3, $+2.8$] n.i.	$+5.10$ [$+2.7$, $+7.1$]
a_{55}	-0.04 [-0.9, $+0.9$] n.i.	$+0.18$ [-0.9, $+1.4$] n.i.	-2.09 [-4.7, -0.3]	$+0.97$ [-1.2, $+3.8$] n.i.
a_{15}	-0.42 [-0.9, $+0.6$] n.i.	-3.12 [-4.8, -0.3] s.o.	-1.45 [-4.3, $+0.5$]	$+0.99$ [-0.2, $+2.3$] n.i.
a_{25}	-0.23 [-0.9, $+0.8$] n.i.	-0.20 [-0.9, $+0.8$] n.i.	$+1.18$ [-0.3, $+2.4$] n.i.	$+1.10$ [-0.2, $+2.3$] n.i.
a_{35}	-0.32 [-0.9, $+0.7$] n.i.	-0.29 [-0.9, $+0.8$] n.i.	$+1.04$ [$+0.3$, $+2.4$]	-6.55 [-13.4, -1.7]
a_{45}	-0.05 [-0.9, $+0.9$] n.i.	$+0.45$ [-0.9, $+1.8$] n.i.	$+2.81$ [-1.3, $+15.1$]	$+1.21$ [-1.9, $+7.5$]

Table S7: Log-evidence of the full and the reduced ($a_{45} = 0$) model at stage (i), mean $\pm$ half-range over three seeds, and $\ln B = \ln \hat{Z}_{\text{full}} - \ln \hat{Z}_{a_{45}=0}$. The last two columns give the a_{45} box and its log-width, which enters $\ln \hat{Z}_{\text{full}}$ as an Occam penalty.

Cond.	$\ln \hat{Z}_{\text{full}}$	$\ln \hat{Z}_{a_{45}=0}$	$\ln B$	a_{45} box	ln(width)
CS	-15.16 ± 0.08	-15.26 ± 0.13	$+0.10$	[-1, 1]	0.69
CH	-9.62 ± 0.12	-9.66 ± 0.00	$+0.05$	[-1, 2]	1.10
DS	-21.48 ± 0.70	-18.81 ± 0.51	-2.67	[-15, 20]	3.56
DH	-21.36 ± 0.38	-23.30 ± 0.34	$+1.94$	[-0.5, 6]	1.87

Table S8: DS: a_{45} by a_{33} mode (stage **[iv]**, three seeds).

Mode	Particles	a_{45} median [5%, 95%]	max log-likelihood
$a_{33} \geq -5$	**[]**	**[+2.4 [+0.5, +4.4]]**	**[]**
$a_{33} < -5$	**[]**	**[+0.8 [−0.2, +1.4]]**	**[]**

Table S9: DH, stage (i): weighted ($\lambda_{\mathrm{Pg}} = 5$, $\lambda_{\mathrm{late}} = 3$) and unweighted likelihood. Ranges over three seeds.

	a_{45} median [5%, 95%]	Pg D21/D15 (measured 2.74)	RMSE
Weighted	+2.9 to +3.2 [+0.5 to +1.1, +5.2 to +5.5]	2.4–2.9	0.058–0.065
Unweighted	+3.0 [−0.06 to +0.17, +5.6]	1.7–3.1	0.059–0.068